Macroeconomic Signals and Tactical Asset Allocation

Historical Evidence on Performance, Persistence, and Portfolio Risk

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Abstract

Can a simple reading of public macroeconomic conditions improve a diversified portfolio? This paper evaluates a monthly rule that reallocates ten percentage points between global equities and intermediate-term Treasuries using corporate profits, Federal Reserve assets, and financial conditions. Historical data vintages are matched to conservative information cutoffs, and portfolio returns include initial entry and drift-aware rebalancing costs. The earlier 2018-2021 validation produces a 1.06 information ratio and a 9.27% annualized return, versus 8.21% for the static benchmark. In the subsequent 46 holding periods from February 2022 to December 2025, the information ratio reverses to -0.48: the rule earns 7.32% annualized, compared with 7.89% for the benchmark. Both comparisons use a one-way cost of ten basis points. Annualized arithmetic active return is -0.51 percentage points; a heteroskedasticity- and autocorrelation-consistent 95% interval ranges from -1.41 to 0.39 points. The rule has the same average target equity weight as the benchmark, but a deeper endpoint drawdown. Its underperformance persists at every cost rate considered. Retrospective attribution locates the principal shortfall in equity underweights during 2023-2024. The reversal raises questions about signal persistence and changing policy conditions; it does not establish a causal regime explanation. Independent verification of portions of the return history remains incomplete.

Keywords: tactical asset allocation; real-time macroeconomic data; portfolio evaluation; transaction costs.

Working paper. This manuscript reports a completed evaluation under stated assumptions, not a peer-reviewed publication. Return-source reconciliation remains open. The author directed the research question; AI tools assisted with implementation and drafting.

1. Introduction

An investor can have a sensible view of the economy and still make a poor allocation decision. Rising corporate profits may indicate stronger business conditions, but equity prices may already reflect the improvement. An expanding central-bank balance sheet may ease financing conditions, or it may be a response to a deterioration that is not yet fully visible. The investment question is therefore not whether macroeconomic conditions matter. It is whether a particular use of information available at the time improves a portfolio relative to leaving its strategic allocation unchanged.

This paper examines that question with a deliberately simple rule. Three public indicators determine whether a seven-fund portfolio increases its equity allocation from 50% to 60%, reduces it to 40%, or remains at its benchmark weights. The corresponding adjustment is made entirely in intermediate-term U.S. Treasuries. The rule is fixed before its reserved-period evaluation, contains no fitted coefficients, and is assessed using historical macroeconomic vintages rather than the latest revised data. Both the rule and its comparators pay transaction costs under the same accounting convention.

The earlier validation is economically encouraging: an information ratio of 1.06 accompanies a compound return of 9.27%, compared with 8.21% for the benchmark. This is the result that motivates a subsequent test. It is not, however, conclusive evidence of persistent skill. At the primary HAC lag, the confidence interval for the early active mean includes zero. The central finding is the contrast between that favorable experience and the later reversal, not a uniformly unsuccessful historical strategy.

The rule does not improve performance in the final evaluated sample. At the primary cost assumption, its annualized compound return is 0.58 percentage points below that of the static allocation. The difference is already negative before costs, so implementation expenses are not the principal explanation. Nor is the result attributable to a permanently lower target equity exposure: the rule averages 50% equity, the same as the benchmark. The relevant economic trade-off is the timing of the equity-versus-Treasury allocation, together with the trading it requires.

The design provides a narrow test rather than a general verdict on macroeconomic investing. It evaluates one signal mapping, one investment universe, and a short historical period. It does not estimate a structural relationship between policy and expected returns, establish causal effects, or compare every plausible allocation model. The paper treats these boundaries as part of the empirical design. A negative result is informative only if the reader can identify the strategy that failed and the evidence on which that conclusion rests.

2. Related literature and research question

The distinction between an appealing predictor and a useful investment decision is central to the return-predictability literature. Welch and Goyal (2008), and the earlier working-paper version by Goyal and Welch (2004), examine the out-of-sample performance of equity-premium predictors against a simple historical benchmark. The present exercise addresses a different object: a discrete portfolio-weight rule rather than an equity-premium forecasting regression. The shared concern is whether information adds value when the comparison is made on returns not used to choose the reported rule. No attempt is made here to reproduce their predictor set or results.

A second issue is the information set. Croushore and Stark (2001) explain the construction and research use of real-time macroeconomic datasets. ALFRED likewise preserves historical vintages that

may differ from current revised observations. A backtest that assigns a subsequently revised value to an earlier decision can give an investor knowledge that was not then available. Historical vintages reduce this problem, but do not eliminate every timing ambiguity: the date on which a vintage is recorded is not necessarily a precise intraday release timestamp.

The interpretation of further experimentation also matters. Harvey, Liu, and Zhu (2016) show why conventional significance thresholds can be misleading when many candidate factors have been examined. Their multiple-testing framework concerns the cross-section of returns and is not imposed mechanically on this portfolio exercise. It nevertheless motivates a clear separation between the fixed evaluation and subsequent exploratory comparisons. Once the reserved returns have been inspected, a different rule evaluated on those same returns is not a new out-of-sample test.

The research question is whether the fixed signal rule produces positive net active returns relative to the strategic allocation. Compound wealth, endpoint drawdown, and the information ratio describe the realized portfolio experience; the mean active return and its interval summarize sampling uncertainty. These are complementary statistics, not interchangeable tests of skill. In particular, a positive difference in compound returns would not by itself identify risk-adjusted alpha, and failure to reject a zero active mean would not demonstrate that the two strategies are economically equivalent.

3. Data and portfolio construction

The investment universe comprises SPY, EFA, EEM, IEF, TIP, GLD, and BIL. The primary benchmark assigns them weights of 30%, 15%, 5%, 25%, 10%, 10%, and 5%, respectively. This allocation combines U.S. and non-U.S. equities with nominal Treasuries, inflation-linked Treasuries, gold, and Treasury bills. The instruments are selected surviving funds. The sample should not be described as a survivorship-free investment universe or as an exhaustive representation of investable assets.

The macro rule uses the year-over-year growth of corporate profits after tax (CPATAX), year-over-year growth of Federal Reserve total assets (WALCL), and the level of the National Financial Conditions Index (NFCI). These variables are interpreted as business profitability, central-bank balance-sheet conditions, and financial conditions. They are proxies, not direct estimates of expected stock returns. Profits can be revised, balance-sheet growth can be endogenous to distress, and financial conditions can overlap with information already reflected in market prices.

For each monthly decision, the feature pipeline uses the historical snapshot eligible at its information cutoff. A vintage is admitted on the following calendar day at midnight in New York; execution then occurs at the first exchange-session close strictly after the cutoff. The convention deliberately avoids assuming immediate intraday access. It remains a date-level reconstruction, however, and should not be interpreted as a complete historical news-release feed. No future observation is used to fill a missing required signal; missing inputs return the allocation to its benchmark weights.

Development and validation were organized around 2010-2017 and 2018-2021 decision vintages. The frozen final evaluation uses January 2022-November 2025 decision vintages, with 47 eligible execution prices defining 46 holding periods. The first entry is February 1, 2022, and the final valuation is December 1, 2025. Annualized statistics use twelve holding periods per year; these endpoints do not

represent four complete calendar years. The final sample was evaluated on November 2, 2026. Historical market narratives were already known, and the sample is no longer untouched.

Portfolio returns use saved vendor-adjusted closes, which incorporate distribution adjustments according to the vendor series. Calendar checks confirm 1,003 daily observations per fund in the 2022-2025 extraction, without missing or extra exchange sessions. That check establishes calendar completeness, not the correctness of the adjustment methodology. Independent return-source reconciliation is incomplete; Appendix B identifies the unresolved cases and their relevance to the research history.

4. Empirical method

Let v_t be the number of favorable signals at decision t. A favorable signal is positive year-over-year profits growth, positive year-over-year Federal Reserve asset growth, or NFCI no greater than zero. The equity tilt, expressed as a fraction of portfolio value, is

$$d_t = 0.10 \text{ if } v_t = 3;\ 0 \text{ if } v_t = 2;\ -0.10 \text{ if } v_t \text{ is } 0 \text{ or } 1. \quad (1)$$

If any required input is unavailable, d_t is zero. Each equity sleeve retains its share of the benchmark equity allocation, so that for i in {SPY, EFA, EEM},

$$w_{i,t} = b_i(0.50 + d_t)/0.50; \quad w_{IEF,t} = 0.25 - d_t. \quad (2)$$

TIP, GLD, and BIL remain at benchmark weights. The weights are nonnegative and sum to one. The strategy neither borrows nor sells short. The comparison therefore concerns a small set of implementable exposure changes rather than unrestricted optimization.

Trading costs are charged on actual purchases and sales, not on changes in target weights alone. Let $p_{i,t}$ denote the pretrade portfolio fraction after the previous holdings have drifted, c the cost per dollar traded, and f_t the fraction of pretrade wealth paid in fees. Post-cost target weights satisfy

$$f_t = c \sum_i |(1 - f_t)w_{i,t} - p_{i,t}|. \quad (3)$$

The implementation solves this fixed-point equation before calculating the holding-period return. If $r_{i,t+1}$ is the adjusted-close return between eligible executions, the net portfolio return is

$$R_{t+1} = (1 - f_t)(1 + \sum_i w_{i,t} r_{i,t+1}) - 1. \quad (4)$$

Each portfolio begins in cash and pays the initial investment cost. Final wealth is marked to market rather than liquidated. The primary scenario sets c to 0.001, or ten basis points; zero, five, and twenty-five basis points are reported as sensitivity cases. These are hypothetical implementation assumptions, not estimates of realized spreads, commissions, taxes, or market impact.

The main comparator is rebalanced to the same strategic weights each period under the same cost convention. A fixed 40% equity allocation and a portfolio of 60% SPY and 40% IEF provide additional exposure comparisons. The former reduces the three equity sleeves proportionally and adds ten percentage points to IEF. The latter changes both geographic composition and diversifying assets; it is not an otherwise-identical higher-equity version of the primary benchmark.

Inference is based on an intercept-only regression of net active returns, $a_t = R_{rule,t} - R_{benchmark,t}$. The covariance estimator uses Bartlett-weighted HAC errors in the tradition of Newey and West (1987), with lag three specified as primary, finite-sample correction, and a Student-t confidence interval. Lag zero and lag six are reported without selecting the most favorable result. With 46 observations, the intervals are descriptive; neither the HAC adjustment nor the choice of a t distribution resolves model-selection risk or limited power.

5. Results: a favorable validation followed by reversal

Table 1 places the earlier validation and the final evaluation on the same footing. Each contains 46 holding periods, uses the same fixed rule and static benchmark, and charges ten basis points per dollar traded. The earlier execution window runs from February 1, 2018 to December 1, 2021; its information ratio is 1.06, with an annualized arithmetic active mean of 1.02 percentage points and tracking error of 0.96%. In the later window, the active mean falls to -0.51 points while tracking error rises modestly to 1.07%. The reversal is primarily a change in average relative performance, not an explosion in tracking error.

The earlier compound return advantage is 1.06 percentage points per year (9.27% versus 8.21%). That numerical similarity to the information ratio is coincidental: compound-return differences and information ratios have different definitions and units. The primary lag-three HAC interval for the early arithmetic active mean is [-0.06, 2.09] percentage points, with $p = 0.063$. The two point estimates are economically different, but their confidence intervals overlap; this paper does not report a formal test of a structural break. Moreover, the earlier validation had already been examined during research development, whereas the final evaluation was reserved until the rule was frozen. The two samples therefore have different selection histories.

Table 1 reports the primary comparison. The macro rule earns 7.32% annualized, against 7.89% for the static allocation. Its information ratio is -0.48 and its maximum endpoint drawdown is -15.97%, compared with -15.14% for the benchmark. Figure 1 traces the corresponding wealth and drawdown paths. Drawdown is measured at holding-period endpoints, so it need not equal the worst daily drawdown an investor would have experienced.

The signal history helps explain what is being compared. Of the 46 implemented decisions, ten increase the equity allocation, ten reduce it, twenty-five leave it neutral, and one uses the missing-input fallback. The average target equity allocation is therefore 50%. The rule has not simply exchanged lower expected returns for a permanently more conservative target allocation. It changes when equity risk is held. Equal average target weights do not imply equal realized beta, volatility, or risk-adjusted exposure, so this observation is not a factor-adjusted attribution.

The rule exceeds the fixed 40% equity comparator, whose annualized return is 6.70%, but that comparison alone offers little evidence of timing ability: the portfolios hold different average target equity exposures. Its return is also close to the 7.30% earned by SPY/IEF 60/40, which has a different asset mix. The principal test remains the comparison against the original seven-fund allocation, not whichever secondary comparator makes the rule look most attractive.

Table 2 reports the uncertainty estimates. At lag three, annualized arithmetic active return is -0.51 percentage points, with a 95% interval of [-1.41, 0.39] and a two-sided p-value of 0.260. The interval

includes zero and modest positive effects, but is centered below zero. This is insufficient evidence of improvement, not proof that the true effect is exactly zero. The arithmetic active mean differs from the 0.58-point gap in annualized compound returns because the two statistics summarize compounding differently.

The negative comparison is not created by the cost assumption. Table 3 and Figure 2 show that the macro rule trails the static allocation at each of the four rates. At zero cost, its annualized return is 7.39%, versus 7.95% for the benchmark. Costs lower both returns and slightly widen the gap. The result is thus unfavorable even under frictionless execution within the otherwise unchanged model.

The annual active-return sums in the frozen record are positive in 2022, negative in 2023 and 2024, and zero in 2025. These are sums of holding-period active returns grouped by exit year, not compound calendar-year performance. The saved weights show that the rule is neutral throughout the holding periods ending in 2025. That final-year equality reflects the absence of an active allocation, rather than a separate period of successful timing.

6. Why might the earlier result fail to persist?

A retrospective accounting decomposition narrows the explanation. Before costs, the active return in each period is the equity tilt multiplied by the return of the proportional global-equity sleeve relative to IEF. The ten risk-on holding periods ending in 2022 contribute a positive 0.22 percentage points to the sum of net active returns. The ten risk-off holding periods ending in 2023-2024 contribute -2.14 points. Neutral and missing-input periods contribute approximately -0.04 points, reflecting residual implementation differences; periods ending in 2025 have no active allocation. These are sums of monthly active returns, not annualized returns or compounded contributions to wealth. The exercise is a post-evaluation diagnostic, not a new independent test.

The vintage feature records identify why the rule became defensive. At every risk-off decision from May 2023 through February 2024, both profits growth and Federal Reserve asset growth are negative, while NFCI remains below zero. Under the majority-vote rule, the two negative growth indicators dominate the favorable financial-conditions reading. The resulting equity underweights lose ground in aggregate because the proportional equity sleeve outperforms IEF over those holding periods. Thus, a generic explanation that the strategy simply failed to protect against the 2022 selloff would misidentify the principal observed source of the shortfall.

One plausible interpretation is a mismatch between backward-looking economic growth rates and forward-looking asset prices. Year-over-year profits can remain negative after investors begin to discount a recovery. A balance-sheet contraction associated with policy normalization may persist even as financial conditions improve. If these mechanisms mattered, the voting system would remain defensive after the expected relative return to equities had improved. The observed disagreement among the signals is consistent with this account, but the study has not identified recovery expectations or a causal policy effect. Testing the account would require a separately specified analysis of release surprises, expectations and the timing of the allocation changes.

A second possibility is that the economic meaning of the balance-sheet variable changed across the two samples. Expansion during a period of monetary accommodation and contraction during a normalization phase need not be symmetric signals for equity-versus-bond returns. Likewise, nominal

Treasuries need not provide the same relative protection in every inflation and policy environment. The present attribution establishes that IEF was the losing alternative to the equity sleeve during the defensive periods in aggregate; it does not establish whether inflation, expected real rates, term premia or earnings expectations caused that relative return. Those channels are competing hypotheses rather than retrospective certainties.

A simpler explanation remains sampling variation and favorable earlier selection. The validation contains only 46 observations, its primary uncertainty interval includes zero, and its average target equity weight is 52.39%, compared with 50.00% in the final evaluation. Part of the favorable early experience could therefore reflect a modest equity overweight, episodic timing success, or both. A risk-adjusted attribution would be needed to separate these channels. The appropriate interpretation of the 1.06 information ratio is promising earlier evidence that did not persist, not a stable population parameter subsequently lost by the calculation.

Subsequent model comparisons belong to a different evidentiary category. Exploratory neural and ridge specifications report higher returns than the frozen rule, but they use history already consumed by the final evaluation. A constant higher-global-equity diagnostic also outperforms the exploratory neural specification. Those results do not establish a machine-learning advantage: model form, exposure, and selection are confounded. They are excluded from the main results tables to avoid presenting a collection of retrospectively examined alternatives as additional confirmatory tests.

Three constraints limit the interpretation. First, 46 observations provide little information about performance across distinct economic environments. Serial dependence and the small number of active decisions further limit what can be inferred from the average. Second, the selected ETF universe and fixed allocation are part of the hypothesis; the result need not extend to other instruments or strategic weights. Third, unresolved source discrepancies mean that exact reproducibility of the saved return series is not the same as independent verification of those returns. Appendix B separates the accounting checks from the remaining data work.

The appropriate next empirical test would use a genuinely new information set or a prospectively specified period, not relabel the existing sample. No prospective return evidence is reported here. Extending the sample, replacing an indicator, or changing a threshold could be useful research, but would constitute a new specification requiring its own rationale and evaluation design.

6.1. Rolling performance and the latest available window

A descriptive stability check uses every available twelve-holding-period window within each saved return block. The earlier research history, spanning 2010-2021 and including the validation observations, supplies 131 overlapping windows: 61 have a positive compound-return difference, 43 a negative difference, and 27 no difference from the benchmark. The final evaluation supplies 35 windows: six positive, 21 negative, and eight with no difference. Figure 3 displays all windows rather than selecting favorable episodes. These counts are not independent trials or estimates of a future success probability; adjacent windows share eleven observations.

The latest available window runs from December 2, 2024 to December 1, 2025. Both portfolios earn 16.69% after the specified trading costs. Active returns and tracking error are zero, making the information ratio undefined. This equality is not evidence that the timing signal recovered: it reflects

benchmark-equivalent positioning. Across final-period windows, the compound-return difference ranges from -2.35 to +0.65 percentage points. The earlier block ranges from -2.70 to +4.34 points. Short-window ratios are particularly unstable when active risk is small; the compound-return difference is therefore reported alongside the ratio.

The diagnostic retains realized costs in the saved continuous returns; it does not restart each rolling window in cash. Windows do not bridge the gap between the two saved blocks, and validation observations are not counted twice. Twelve holding periods are approximately, rather than exactly, one calendar year. No 2026 performance is reported: the checked execution-price export ends in December 2025. These retrospective comparisons extend the description of the existing evidence, not the independent test sample.

7. Conclusion

The earlier validation provides a clear positive result in realized performance: a 1.06 information ratio and a compound-return advantage over the static allocation. The subsequent test reverses that result. For the specified portfolio and final historical period, the three-indicator allocation rule does not improve on maintaining the strategic weights. It produces a lower compound return, a negative information ratio, and a deeper endpoint drawdown. These differences persist without transaction costs. The uncertainty interval does not exclude a modest benefit, but the observed evidence provides no affirmative reason to prefer the rule to the benchmark. The result supports retaining the static comparator in this experiment, while leaving the broader question of macroeconomic allocation open.

The reversal motivates a distinction between signal instability and model adaptation. The next extension should carry the original rule forward unchanged and compare it with a separately specified adaptive alternative using only information available before each decision. A revision informed by the observed reversal would remain exploratory even under walk-forward estimation; its stronger test would require subsequently observed returns. The earlier validation establishes that the rule delivered economically meaningful gains in one historical setting, while the subsequent attribution identifies a specific weakness: defensive signals persisted as financial conditions improved. Together, these findings sharpen the next research question from whether macroeconomic information matters to when its interpretation should change. The contribution is therefore a transparent empirical foundation for studying more adaptable allocation rules, with explicit information timing, trading costs and benchmark comparisons. Whether adaptation improves performance remains to be tested, but the evidence now provides a concrete and economically motivated direction for that work.

References

Croushore, D., and T. Stark. 2001. A real-time data set for macroeconomists. *Journal of Econometrics* 105:111-130.

Federal Reserve Bank of St. Louis. n.d. ALFRED: Archival Federal Reserve Economic Data, Help. https://alfred.stlouisfed.org/help. Accessed November 5, 2026.

Goyal, A., and I. Welch. 2004. A comprehensive look at the empirical performance of equity premium prediction. NBER Working Paper 10483. https://doi.org/10.3386/w10483.

Harvey, C. R., Y. Liu, and H. Zhu. 2016. ... and the cross-section of expected returns. *Review of Financial Studies* 29:5-68. https://doi.org/10.1093/rfs/hhv059.

Newey, W. K., and K. D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55:703-708.

Welch, I., and A. Goyal. 2008. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies* 21:1455-1508. https://doi.org/10.1093/rfs/hhm014.

Table 1. Earlier validation and final evaluation

Measure	Earlier validation	Final evaluation
Execution window	Feb. 2018-Dec. 2021	Feb. 2022-Dec. 2025
Holding periods	46	46
Rule CAGR (%)	9.27	7.32
Benchmark CAGR (%)	8.21	7.89
Information ratio	1.06	-0.48
Active mean (pp/year)	1.02	-0.51
Tracking error (%/year)	0.96	1.07
HAC(3) 95% interval (pp/year)	[-0.06, 2.09]	[-1.41, 0.39]
HAC(3) p-value	0.063	0.260
Average target equity (%)	52.39	50.00

Notes: Identical fixed rule, benchmark and ten-basis-point cost convention. Both windows start in cash and pay initial entry costs. Earlier validation had been examined before the final rule freeze; the final sample was reserved until evaluation. Intervals concern the arithmetic active mean, not the information ratio. No formal cross-period break test is reported. Sources: saved validation and final-evaluation outputs.

Table 1 (continued). Final-period portfolio comparison

Portfolio	Compound return (%)	Maximum drawdown (%)	Information ratio
Macro rule	7.32	-15.97	-0.48
Static allocation	7.89	-15.14	--
Fixed 40% equity	6.70	-14.32	-0.95
SPY/IEF 60/40	7.30	-15.80	-0.13

Notes: 46 holding periods, February 1, 2022-December 1, 2025. Compound return is annualized using 12 holding periods per year. Costs apply to purchases and sales, including initial deployment; no terminal liquidation is charged. Drawdown uses holding-period endpoints. Information ratios are relative to the static allocation and use square-root-of-12 annualization. -- denotes not applicable. Source: frozen evaluation; adjusted-close input reconciliation is incomplete.

Table 2. Uncertainty in the mean active return

HAC lag	Active mean (pp/year)	Standard error (pp/year)	95% interval (pp/year)	p-value
0	-0.51	0.55	[-1.61, 0.59]	0.355
3 (primary)	-0.51	0.45	[-1.41, 0.39]	0.260
6	-0.51	0.42	[-1.36, 0.33]	0.230

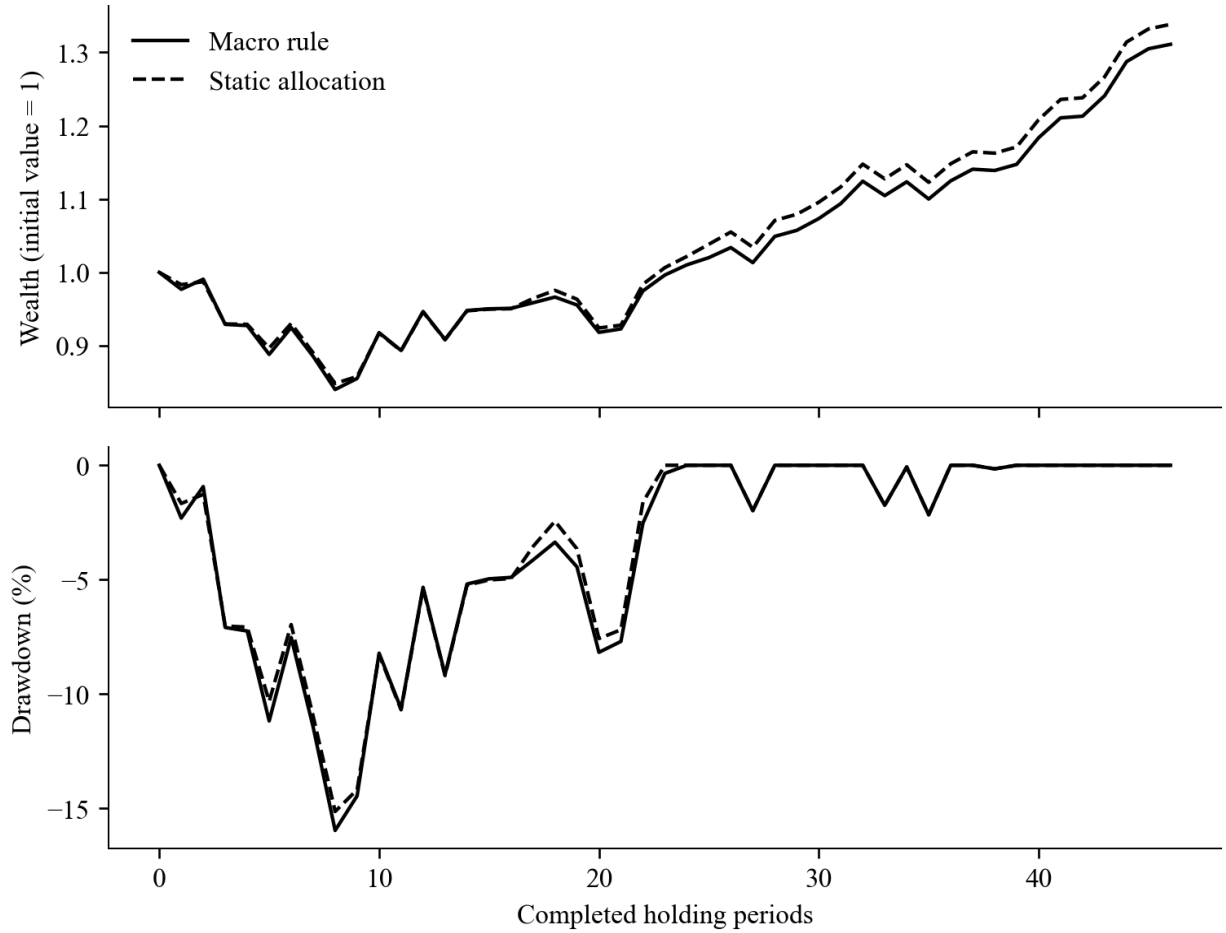
Notes: Active return is the rule minus the static benchmark at ten-basis-point costs. Estimates are from an intercept-only regression with Bartlett HAC covariance, finite-sample correction and Student-t inference. n = 46. Annualization multiplies the monthly mean and standard error by 12. pp = percentage points. These intervals are not adjusted for model selection.

Table 3. Transaction-cost sensitivity

Cost (bps)	Macro rule CAGR (%)	Static CAGR (%)	Difference (pp)
0	7.39	7.95	-0.56
5	7.35	7.92	-0.57
10	7.32	7.89	-0.58
25	7.22	7.82	-0.60

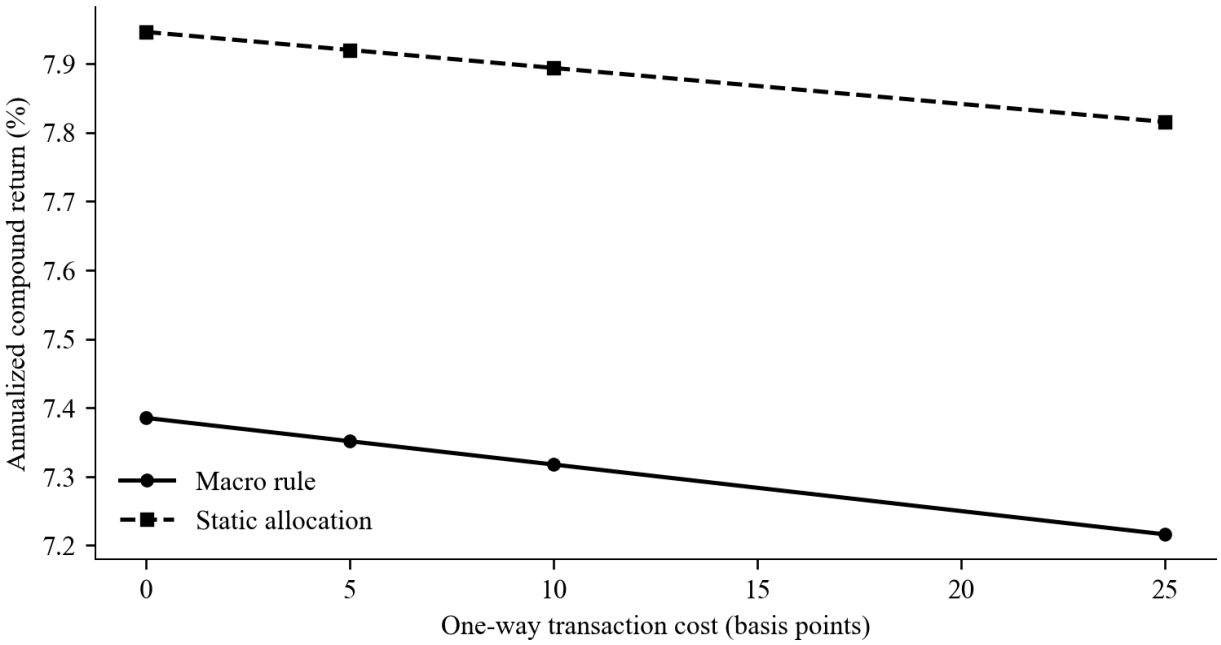
Notes: All cost scenarios use the identical rule, observations and execution dates. Costs are hypothetical one-way rates per dollar traded. The final column is the difference in annualized compound returns, not the annualized arithmetic active mean. Source: frozen evaluation.

Figure 1. Wealth and drawdown



Notes: Net holding-period returns at ten-basis-point costs. Both portfolios start at one before the initial investment cost. The horizontal axis counts completed holding periods; it is not a calendar-month index. Drawdown is relative to the running maximum including initial wealth. Solid: macro rule. Dashed: static allocation. Source: saved final-evaluation return export.

Figure 2. Compound returns under alternative trading costs



Notes: Each plotted value is an annualized compound return from the frozen evaluation, not a fitted prediction. The rule and benchmark use the same cost rate and post-cost accounting. Solid circles: macro rule. Dashed squares: static allocation. The vertical axis is intentionally restricted to show the comparison; it is not a bar chart.

Appendix A. Return and inference definitions

With net holding-period returns R_1, ..., R_T, wealth is $W_0 = 1$ and $W_t = W_{t-1}(1 + R_t)$. Annualized compound return is $W_T^{12/T} - 1$. Maximum drawdown is the minimum of $W_t/\max(W_0, ..., W_t) - 1$. Including W_0 ensures that any initial-entry loss is counted.

For active returns a_t, the annualized arithmetic mean is 12 times the sample mean. The information ratio is the sample mean divided by the sample standard deviation (with T - 1 degrees of freedom), multiplied by the square root of 12. The ratio does not incorporate a serial-correlation adjustment; the separately reported HAC interval addresses uncertainty in the mean, not an adjusted information ratio.

Pretrade fractions for the next rebalance are obtained by marking the existing holdings to the next execution price and dividing by gross portfolio wealth. Transaction costs reduce wealth before the holding return. A separate share-and-dollar ledger, using a bracketed fee solver, reproduces the rule and benchmark across all four cost rates within 6×10^{-16} in the checked accounting outputs. This establishes numerical agreement between implementations conditional on the same inputs.

Appendix B. Data qualifications and reproducibility

The frozen protocol records the rule, evaluation window, benchmarks, cost cases, inference settings, and hashes of the governing code and features. The original result is retained. Reproduction compares against that result; it neither changes the rule nor restores the evaluated period to holdout status. The presentation in this manuscript reads those saved outputs and does not fit a model.

The broader macro archive contains 1,070 of 1,152 requested historical snapshots. These are archive-coverage counts across six series, not the number of available portfolio decisions. The final rule uses three signals and invokes a benchmark fallback once in its implemented sample. The distinction matters: aggregate archive completeness cannot be used to infer the absence of missing observations in a particular strategy, and a successful strategy run does not certify the entire archive.

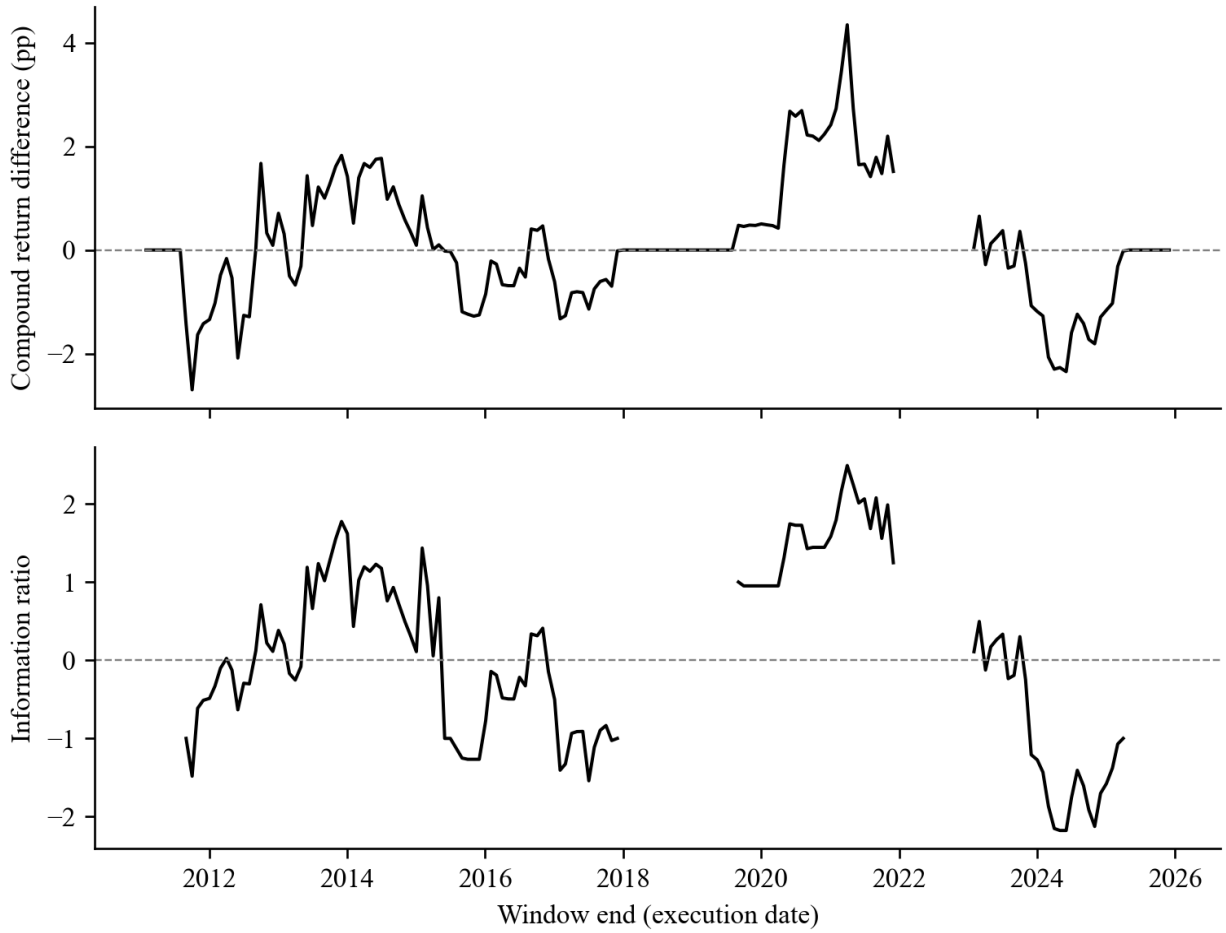
The independent reconciliation register records 16,121 Nasdaq/Yahoo daily comparisons and retains two unresolved official-close cases: SPY on July 16, 2019, and IEF on July 9, 2020. It also retains 228 older GLD conflicts, missing early independent history for 1,721 sessions per fund, and two BIL distribution cases dated March 1 and September 1, 2022 requiring original-source adjudication. Annual return-methodology residuals remain for IEF in 2024 and TIP in 2022. These items differ in timing and potential impact; the counts should not be added together as independent errors in the final sample.

The 2019 and 2020 price cases predate the reserved evaluation but affect the development and validation evidence chain. The BIL cases and the annual residuals overlap the final period and require separate attention. A retained-cash sensitivity calculation changes the rule CAGR by approximately -0.0352 basis points per year and leaves the negative comparison intact. Because that calculation shares closing prices with the original construction, it does not settle independent price accuracy or verify every distribution adjustment.

Two commands in the private project environment reproduce the main checks: *python final_evaluation.py --reproduce* and *python verify_final_accounting.py*. Both completed successfully on November 5, 2026. The manuscript package includes tables, figures, the frozen protocol, verification records, and hashes identifying the underlying evidence. It is not a standalone redistribution of every raw record needed to rebuild the historical database. A complete independent replication requires access to that archive and its input conventions.

The literature cited here motivates the design and interpretation. It is not a systematic claim that this exact combination of indicators is novel. Before external submission, the source exceptions and the incremental contribution relative to the closest allocation studies must be resolved. The conclusions in Section 7 remain conditional on the documented return inputs.

Figure 3. Stability across twelve-holding-period windows



Notes: All 166 overlapping windows in the saved 2010-2021 research block and 2022-2025 final block. Upper panel: strategy compound return minus benchmark compound return, in percentage points. Lower panel: annualized information ratio, omitted where tracking error is zero (35 windows). A zero tracking error makes the ratio undefined, not zero. Lines do not join the separate blocks. Costs are inherited from the saved return path rather than recalculated for fresh investment at each window start. These are retrospective, dependent observations, not additional independent tests. Source: saved ten-basis-point returns and execution dates; reproducible diagnostic exports accompany the manuscript.